

ATHARV C. JOSHI

AI/ML Engineer — Robotics — Computer Vision — GenAI Developer

📍 Burhanpur, Madhya Pradesh 📞 +91 93025 73730 ✉️ joshiatharv348@gmail.com
🐙 github.com/atharv348 🌐 LinkedIn 🌐 Portfolio

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, HTML, CSS, SQL
- **AI/ML & GenAI:** PyTorch, TensorFlow, OpenCV, NLP, Computer Vision, RAG, LangChain, ChromaDB, LLMs
- **Robotics & IoT:** ROS2, Raspberry Pi, ESP32, Sensor Fusion, Autonomous Navigation, Gazebo
- **Backend & Cloud:** FastAPI, REST APIs, SQLAlchemy, Docker, AWS EC2, PostgreSQL
- **Frontend & Tools:** React.js, Tailwind CSS, Git, Linux, Vercel, Render

PROJECTS

Autonomous Surveillance Rover ROS2, OpenCV, Raspberry Pi

- Built an autonomous rover on Linux using ROS2 navigation stack with obstacle avoidance, sensor fusion, and real-time path planning for autonomous indoor/outdoor surveillance.
- Developed OpenCV-based perception modules for terrain and obstacle detection, improving navigation stability and reducing manual intervention during multiple test runs.
- Simulated and validated rover workflows in Gazebo before Raspberry Pi deployment, enabling faster debugging cycles and efficient edge-based execution.

SvasthaAI – AI-Powered Health OS Ecosystem FastAPI, LangChain, AWS

🐙 GitHub Repo

- Developed a multi-organ disease screening platform supporting 5+ clinical domains using PyTorch and OpenCV for image-based medical inference.
- Built a RAG pipeline using LangChain, ChromaDB, and Llama-3 achieving 90%+ contextual retrieval accuracy for medical PDF analysis and multilingual wellness coaching.
- Designed and deployed Dockerized backend services using FastAPI, SQLAlchemy, and AWS EC2 with JWT authentication, supporting concurrent multi-user API requests.

AI Legal Intelligence System (ALIS) Legal-BERT, OCR, React

🐙 GitHub Repo

- Developed a LegalTech platform for semantic legal search, OCR-based document analysis, and AI-powered legal assistance using transformer-based NLP models.
- Implemented multilingual NLP pipelines and OCR workflows across large-scale legal datasets, improving legal document retrieval and contextual search efficiency.
- Built automated legal drafting and chatbot support using FastAPI and React, reducing manual effort for generating standard legal templates and responses.

IoT-Based Smart Monitoring Systems ESP32, Raspberry Pi, IoT Sensors

- Engineered IoT-based monitoring solutions for water quality analysis, smart irrigation, and laser-based perimeter security using ESP32 and Raspberry Pi.
- Integrated multiple sensor arrays with cloud dashboards for real-time telemetry, automated alert generation, and remote monitoring capabilities.
- Developed rule-based actuator automation and real-time event logging workflows, improving monitoring efficiency and reducing manual supervision requirements.

EDUCATION

D. Y. Patil Agriculture and Technical University, Kolhapur May 2027

B.Tech in Artificial Intelligence & Machine Learning

Current CGPA: 8.32 / 10

Padmashree Dr. Vitthalrao Vikhe Patil Sainik School 2023

Class XII – 63%

ACHIEVEMENTS

- 2nd Place – Rover Domain, MindSpark 2025 (COEP Pune)
- 13th Rank Nationwide – RoboRun, Shastra 2026 (IIT Madras)
- Cyber Suraksha Warrior of the Year 2025 – Quick Heal Foundation

CORE COMPETENCIES

Team Leadership • Problem Solving • System Design • Technical Communication • Research & Development

LANGUAGES

English (Basic) Hindi (Native) Marathi (Native)